

Package ‘manMetaVAR’

March 14, 2026

Title Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients

Version 0.9.8

Description Research compendium for the manuscript
Pesigan, I. J. A., et al. (In Preparation).
Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients.
<doi:10.0000/0000000000>.

URL <https://github.com/jeksterslab/manMetaVAR>,
<https://jeksterslab.github.io/manMetaVAR/>,
<https://osf.io/uenr7/>, <https://doi.org/10.0000/0000000000>

BugReports <https://github.com/jeksterslab/manMetaVAR/issues>

License MIT + file LICENSE

Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 4.5.0), OpenMx (>= 2.22.11), fitVARMxID (>= 1.0.2),
metaDyn (>= 1.0.0)

Imports stats, utils, simStateSpace (>= 1.2.15), lavaan (>= 0.6-21),
ggplot2

Remotes jeksterslab/fitVARMxID, jeksterslab/metaDyn

Suggests knitr, rmarkdown, testthat

RoxygenNote 7.3.3.9000

NeedsCompilation no

Author Ivan Jacob Agaloos Pesigan [aut, cre, cph] (ORCID:
<<https://orcid.org/0000-0003-4818-8420>>)

Maintainer Ivan Jacob Agaloos Pesigan <r.jeksterslab@gmail.com>

Contents

Check	2
CheckFitDTVAR	3
CheckFitMetaVAR	4
CheckFitMplus	5
CheckFitNaive	5
Compress	6
FigBias	7
FigCoverage	8
FigPower	9
FigRMSE	10
FitDTVAR	11
FitMetaVAR	11
FitMplus	12
FitNaive	13
GenData	14
manmetavar-data-methods	15
manmetavar-dtvar-methods	15
manmetavar-metavar-methods	17
manmetavar-mplus-methods	19
model	20
MplusInput	21
params	22
results	22
Sim	24
SimFitDTVAR	25
SimFitMetaVAR	26
SimFitMplus	27
SimFitNaive	28
SimFN	29
SimGenData	29
SimProj	30
Sum	30
SumFitMetaVARNormal	32
SumFitMetaVARRobust	32
SumFitMplus	33
SumFitNaive	34

Index

35

Check	<i>Check Replication</i>
-------	--------------------------

Description

Check Replication

Usage

```
Check(taskid, repid, output_folder, naive, metavar, mplus)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitDTVAR	<i>Check Replication - FitDTVAR</i>
---------------	-------------------------------------

Description

Check Replication - FitDTVAR

Usage

```
CheckFitDTVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of manCTMed:::SimSuffix().

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMetaVAR	<i>Check Replication - FitMetaVAR</i>
-----------------	---------------------------------------

Description

Check Replication - FitMetaVAR

Usage

```
CheckFitMetaVAR(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitMplus	<i>Check Replication - FitMplus</i>
---------------	-------------------------------------

Description

Check Replication - FitMplus

Usage

```
CheckFitMplus(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

CheckFitNaive	<i>Check Replication - FitNaive</i>
---------------	-------------------------------------

Description

Check Replication - FitNaive

Usage

```
CheckFitNaive(taskid, repid, output_folder, suffix)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .

Details

This function is executed via the Check function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

Compress

Compress Replication

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

FigBias*Plot Relative Bias*

Description

Plot relative bias for the fixed and random effects.

Usage

```
FigBias(  
  results,  
  bias = FALSE,  
  method = c("BMLVAR", "MetaVAR", "Naive"),  
  parameters = "both",  
  ylim = c(-0.15, 0.15),  
  x_lab_size = 6  
)
```

Arguments

results	results data frame.
bias	Logical. If bias = TRUE, plot absolute bias. If bias = FALSE, plot relative bias. Note that when parameter value is equal to zero, absolute bias is used instead of relative bias.
method	Character vector. Methods to include in the plot.
parameters	Character string. Parameters to plot. Valid values include "fixed", "random", and "both".
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.
x_lab_size	Numeric. x-axis label size.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigCoverage\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

Examples

```
## Not run:  
data(results, package = "manMetaVAR")  
FigBias(results)  
  
## End(Not run)
```

FigCoverage*Plot Coverage Probabilities*

Description

Plot coverage probabilities for the fixed and random effects.

Usage

```
FigCoverage(  
  results,  
  method = c("BMLVAR", "MetaVAR", "Naive"),  
  parameters = "both",  
  ylim = c(0.5, 1),  
  x_lab_size = 6  
)
```

Arguments

results	results data frame.
method	Character vector. Methods to include in the plot.
parameters	Character string. Parameters to plot. Valid values include "fixed", "random", and "both".
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.
x_lab_size	Numeric. x-axis label size.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigPower\(\)](#), [FigRMSE\(\)](#)

Examples

```
## Not run:  
data(results, package = "manMetaVAR")  
FigCoverage(results)  
  
## End(Not run)
```


Description

Plot statistical power probabilities for the fixed and random effects.

Usage

```
FigPower(  
  results,  
  method = c("BMLVAR", "MetaVAR", "Naive"),  
  parameters = "both",  
  ylim = c(0, 1),  
  x_lab_size = 7.5  
)
```

Arguments

results	results data frame.
method	Character vector. Methods to include in the plot.
parameters	Character string. Parameters to plot. Valid values include "fixed", "random", and "both".
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.
x_lab_size	Numeric. x-axis label size.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigRMSE\(\)](#)

Examples

```
## Not run:  
data(results, package = "manMetaVAR")  
FigPower(results)  
  
## End(Not run)
```

FigRMSE

*Plot RMSE***Description**

Plot RMSE for the fixed and random effects.

Usage

```
FigRMSE(
  results,
  method = c("BMLVAR", "MetaVAR", "Naive"),
  parameters = "both",
  ylim = c(0, 0.25),
  x_lab_size = 6
)
```

Arguments

results	results data frame.
method	Character vector. Methods to include in the plot.
parameters	Character string. Parameters to plot. Valid values include "fixed", "random", and "both".
ylim	Numeric vector of length 2. Lower and upper limits for the y-axis.
x_lab_size	Numeric. x-axis label size.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: [FigBias\(\)](#), [FigCoverage\(\)](#), [FigPower\(\)](#)

Examples

```
## Not run:
data(results, package = "manMetaVAR")
FigRMSE(results)

## End(Not run)
```

FitDTVAR

*Fit the Model using the fitVARMxID Package***Description**

The function fits the model using the [fitVARMxID](#) package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR

*Multivariate Meta-Analysis using the metaDyn Package***Description**

The function performs multivariate meta-analysis using the [metaDyn](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

FitMplus

Fit the Model using Mplus

Description

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 40000L,
  fscores = NULL,
  plot = FALSE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL,
  seed = NULL
)
```

Arguments

data	R object. Output of the GenData() function.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitNaive\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitMplus(data = data)
print(fit)
summary(fit)

## End(Not run)
```

FitNaive

Naive

Description

The function performs the naive “fit-many-then-summarize”.

Usage

```
FitNaive(fit)
```

Arguments

fit	R object. Output of the FitDTVAR() function.
-----	--

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitNaive(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

GenData

Simulate Data

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid, seed = NULL)
```

Arguments

taskid	Positive integer. Task ID.
seed	Integer. Random seed.

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

`manmetavar-data-methods`*Methods for Objects of Class `manmetavar.data`*

Description

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)
```

```
## S3 method for class 'manmetavar.data'
summary(object, ...)
```

```
## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

<code>x</code>	Object of class <code>manmetavar.data</code> .
<code>...</code>	additional arguments.
<code>object</code>	Object of class <code>manmetavar.data</code> .

Author(s)

Ivan Jacob Agaloos Pesigan

`manmetavar-dtvar-methods`*Methods for Objects of Class `manmetavar.dtvar`*

Description

This page documents the available methods for objects of class `manmetavar.dtvar`.

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
```

```
    nu = TRUE,
    psi = TRUE,
    theta = TRUE,
    ncores = NULL,
    ...
)

## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
```


...
)

Arguments

object	Object of class <code>manmetavar.dtvvar</code> .
mu	Logical. If <code>mu = TRUE</code> , include estimates of the <code>mu</code> vector, if available. If <code>mu = FALSE</code> , exclude estimates of the <code>mu</code> vector.
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the <code>alpha</code> vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the <code>alpha</code> vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the <code>beta</code> matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the <code>beta</code> matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the <code>nu</code> vector, if available. If <code>nu = FALSE</code> , exclude estimates of the <code>nu</code> vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the <code>psi</code> matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the <code>psi</code> matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the <code>theta</code> matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the <code>theta</code> matrix.
ncores	Positive integer. Number of cores to use.
...	additional arguments.
robust	Logical. If <code>TRUE</code> , use robust (sandwich) sampling variance-covariance matrix. If <code>FALSE</code> , use normal theory sampling variance-covariance matrix.
x	Object of class <code>manmetavar.dtvvar</code> .
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-metavar-methods

Methods for Objects of Class `manmetavar.metavar`

Description

This page documents the available methods for objects of class `manmetavar.metavar`.

This page documents the available methods for objects of class `manmetavar.naive`.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)

## S3 method for class 'manmetavar.naive'
coef(object, ...)

## S3 method for class 'manmetavar.naive'
vcov(object, ...)

## S3 method for class 'manmetavar.naive'
print(x, ...)

## S3 method for class 'manmetavar.naive'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

object	Object of class <code>manmetavar.naive</code> .
...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available.
x	Object of class <code>manmetavar.naive</code> .
alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mplus-methods

Methods for Objects of Class manmetavar.mplus

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

<code>object</code>	Object of class <code>manmetavar.mplus</code> .
<code>median</code>	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.
<code>burnin</code>	Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.
<code>...</code>	additional arguments.
<code>alpha</code>	Numeric vector. Significance level α .
<code>digits</code>	Integer indicating the number of decimal places to display.
<code>parm</code>	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.

level	the confidence level required.
x	Object of class <code>manmetavar.mplus</code> .
what	Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.
legend_loc	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

model	<i>Model Parameters</i>
-------	-------------------------

Description

Model Parameters

Usage

```
data(model)
```

Format

A list with 15 elements:

k Number of variables.

mu_mu Mean of the set-point vector μ .

mu_sigma Covariance matrix of the parameter μ .

mu_sigma_l Cholesky factor of the covariance matrix of the parameter μ .

beta_mu Mean of lagged coefficients matrix β .

beta_sigma Covariance matrix of the parameter $\text{vec}(\beta)$.

beta_sigma_l Cholesky factor of the covariance matrix of the parameter $\text{vec}(\beta)$.

psi Process noise covariance matrix Ψ .

psi_l Cholesky factor of the process noise covariance matrix Ψ .

psi_d_ldl uc_d of the LDL' decomposition of the process noise covariance matrix Ψ . See [fitVARMxID::LDL\(\)](#).

psi_l_ldl s_l of the LDL' decomposition of the process noise covariance matrix Ψ . See [fitVARMxID::LDL\(\)](#).

ma_fixed Vector of fixed effects $\theta = [\mu, \text{vec}(\beta)]'$

ma_random Matrix of random effects $\theta = [\mu, \text{vec}(\beta)]'$

ma_random_d_ldl uc_d of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

ma_random_l_ldl s_l of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

Author(s)

Ivan Jacob Agaloos Pesigan

MplusInput	<i>Create Mplus Input File</i>
------------	--------------------------------

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [FitNaive\(\)](#)

Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
```

```
    chains = 2L,  
    iter = 40000L,  
    fscores = 1000L,  
    plot = TRUE,  
    ncores = NULL,  
    seed = 42  
  )  
)
```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

data(params)

Format

A dataframe with 12 rows and 3 columns:

- taskid** Simulation Task ID.
- n** Sample size.
- time** Number of measurement occasions. If NA, measurement occasions vary across IDs.

Author(s)

Ivan Jacob Agaloos Pesigan

results	<i>Simulation Results</i>
---------	---------------------------

Description

Simulation Results

Usage

data(results)

Format

A dataframe with I rows and J columns:

taskid Simulation Task ID.

replications Number of replications.

parnames Parameter names.

parameter Population parameter value.

method Method used to fit the data.

n Sample size.

time Number of measurement occasions.

ci Confidence/credible intervals method.

est Mean parameter estimate.

se Mean standard error.

t Mean test statistic.

p Mean p -value.

ll Mean lower limit of the 95% confidence/credible interval.

ul Mean upper limit of the 95% confidence/credible interval.

sig Proportion of statistically significant results.

theta_hit Proportion of replications where the confidence intervals contained the population parameter.

sq_error Mean squared error.

bias Bias.

rel_bias Relative bias.

se_bias Bias in standard error estimate.

rel_se_bias Relative bias in standard error estimate.

rmse Root mean square error.

coverage Coverage probability.

power Statistical power.

par_idx Parameter index.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim

*Simulation Replication***Description**

Simulation Replication

Usage

```

Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  naive,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)

```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.
naive	Logical. If naive = TRUE, fit the naive approach.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR	<i>Simulation Replication - FitMetaVAR</i>
---------------	--

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus

Simulation Replication - FitMplus

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitNaive

Simulation Replication - FitNaive

Description

Simulation Replication - FitNaive

Usage

```
SimFitNaive(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-mplus".
output_folder	Character string. Output folder.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .

Value

Returns a character string file name with the `output_folder` in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj	<i>Simulation Project Name</i>
---------	--------------------------------

Description

Simulation Project Name

Usage

SimProj()

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	<i>Summary</i>
-----	----------------

Description

Summary

Usage

```
Sum(  
  taskid,  
  reps,  
  output_folder,  
  overwrite,  
  integrity,  
  naive,  
  metavar_normal,  
  metavar_robust,  
  mplus,  
  ncores  
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
naive	Logical. If naive = TRUE, summarize naive estimates.
metavar_normal	Logical. If metavar_normal = TRUE, summarize metaVAR model using normal confidence intervals.
metavar_robust	Logical. If metavar_robust = TRUE, summarize metaVAR model using robust (sandwich) confidence intervals.
mplus	Logical. If mplus = TRUE, summarize the DSEM model.
ncores	Positive integer. Number of cores to use.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal *Summary: Normal Theory (FitMetaVAR)*

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARRobust *Summary: Robust (FitMetaVAR)*

Description

Summary: Robust (FitMetaVAR)

Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```


Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus	<i>Summary (FitMplus)</i>
-------------	---------------------------

Description

Summary (FitMplus)

Usage

SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitNaive	<i>Summary: Naive</i>
-------------	-----------------------

Description

Summary: Naive

Usage

SumFitNaive(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

Index

- * **Compression Functions**
 - Compress, [6](#)
- * **Data Generation Functions**
 - GenData, [14](#)
- * **Figure Functions**
 - FigBias, [7](#)
 - FigCoverage, [8](#)
 - FigPower, [9](#)
 - FigRMSE, [10](#)
- * **Model Fitting Functions**
 - FitDTVAR, [11](#)
 - FitMetaVAR, [11](#)
 - FitMplus, [12](#)
 - FitNaive, [13](#)
 - MplusInput, [21](#)
- * **check**
 - Check, [2](#)
 - CheckFitDTVAR, [3](#)
 - CheckFitMetaVAR, [4](#)
 - CheckFitMplus, [5](#)
 - CheckFitNaive, [5](#)
- * **compress**
 - Compress, [6](#)
- * **data**
 - model, [20](#)
 - params, [22](#)
 - results, [22](#)
- * **figure**
 - FigBias, [7](#)
 - FigCoverage, [8](#)
 - FigPower, [9](#)
 - FigRMSE, [10](#)
- * **fit**
 - FitDTVAR, [11](#)
 - FitMplus, [12](#)
 - MplusInput, [21](#)
 - SimFitDTVAR, [25](#)
 - SimFitMetaVAR, [26](#)
 - SimFitMplus, [27](#)
 - SimFitNaive, [28](#)
- SimFitNaive, [28](#)
- * **gendata**
 - GenData, [14](#)
 - SimGenData, [29](#)
- * **manCTMed**
 - SimFN, [29](#)
- * **manMetaVAR**
 - Check, [2](#)
 - CheckFitDTVAR, [3](#)
 - CheckFitMetaVAR, [4](#)
 - CheckFitMplus, [5](#)
 - CheckFitNaive, [5](#)
 - Compress, [6](#)
 - FigBias, [7](#)
 - FigCoverage, [8](#)
 - FigPower, [9](#)
 - FigRMSE, [10](#)
 - FitDTVAR, [11](#)
 - FitMetaVAR, [11](#)
 - FitMplus, [12](#)
 - FitNaive, [13](#)
 - GenData, [14](#)
 - MplusInput, [21](#)
 - Sim, [24](#)
 - SimFitDTVAR, [25](#)
 - SimFitMetaVAR, [26](#)
 - SimFitMplus, [27](#)
 - SimFitNaive, [28](#)
 - SimGenData, [29](#)
 - SimProj, [30](#)
 - Sum, [30](#)
 - SumFitMetaVARNormal, [32](#)
 - SumFitMetaVARRobust, [32](#)
 - SumFitMplus, [33](#)
 - SumFitNaive, [34](#)
- * **meta**
 - FitMetaVAR, [11](#)
 - FitNaive, [13](#)
- * **methods**

- manmetavar-data-methods, 15
- manmetavar-dtvar-methods, 15
- manmetavar-metavar-methods, 17
- manmetavar-mplus-methods, 19
- * **parameters**
 - model, 20
 - params, 22
- * **results**
 - results, 22
- * **simulation**
 - Check, 2
 - CheckFitDTVAR, 3
 - CheckFitMetaVAR, 4
 - CheckFitMplus, 5
 - CheckFitNaive, 5
 - Sim, 24
 - SimFitDTVAR, 25
 - SimFitMetaVAR, 26
 - SimFitMplus, 27
 - SimFitNaive, 28
 - SimFN, 29
 - SimGenData, 29
 - SimProj, 30
 - Sum, 30
 - SumFitMetaVARNormal, 32
 - SumFitMetaVARRobust, 32
 - SumFitMplus, 33
 - SumFitNaive, 34
- * **summary**
 - Sum, 30
 - SumFitMetaVARNormal, 32
 - SumFitMetaVARRobust, 32
 - SumFitMplus, 33
 - SumFitNaive, 34
- Check, 2
- CheckFitDTVAR, 3
- CheckFitMetaVAR, 4
- CheckFitMplus, 5
- CheckFitNaive, 5
- coef.manmetavar.dtvvar
 - (manmetavar-dtvar-methods), 15
- coef.manmetavar.metavar
 - (manmetavar-metavar-methods), 17
- coef.manmetavar.mplus
 - (manmetavar-mplus-methods), 19
- coef.manmetavar.naive
 - (manmetavar-metavar-methods), 17
- Compress, 6
- confint.manmetavar.metavar
 - (manmetavar-metavar-methods), 17
- confint.manmetavar.mplus
 - (manmetavar-mplus-methods), 19
- FigBias, 7
- FigBias(), 8–10
- FigCoverage, 8
- FigCoverage(), 7, 9, 10
- FigPower, 9
- FigPower(), 7, 8, 10
- FigRMSE, 10
- FigRMSE(), 7–9
- FitDTVAR, 11
- FitDTVAR(), 12, 13, 21
- FitMetaVAR, 11
- FitMetaVAR(), 11, 13, 21
- FitMplus, 12
- FitMplus(), 11–13, 21
- FitNaive, 13
- FitNaive(), 11–13, 21
- fitVARMxID, 11
- fitVARMxID::LDL(), 20
- GenData, 14
- GenData(), 11, 13
- manmetavar-data-methods, 15
- manmetavar-dtvar-methods, 15
- manmetavar-metavar-methods, 17
- manmetavar-mplus-methods, 19
- metaDyn, 11
- model, 20
- MplusInput, 21
- MplusInput(), 11–13
- params, 22
- plot.manmetavar.data
 - (manmetavar-data-methods), 15
- plot.manmetavar.mplus
 - (manmetavar-mplus-methods), 19
- print.manmetavar.data
 - (manmetavar-data-methods), 15
- print.manmetavar.dtvvar
 - (manmetavar-dtvar-methods), 15

`print.manmetavar.metavar`
 (`manmetavar-metavar-methods`),
 17

`print.manmetavar.naive`
 (`manmetavar-metavar-methods`),
 17

`results`, 22

`Sim`, 24

`SimFitDTVAR`, 25

`SimFitMetaVAR`, 26

`SimFitMplus`, 27

`SimFitNaive`, 28

`SimFN`, 29

`SimGenData`, 29

`SimProj`, 30

`simStateSpace::SimSSMIVary()`, 14

`Sum`, 30

`SumFitMetaVARNormal`, 32

`SumFitMetaVARRobust`, 32

`SumFitMplus`, 33

`SumFitNaive`, 34

`summary.manmetavar.data`
 (`manmetavar-data-methods`), 15

`summary.manmetavar.dtvvar`
 (`manmetavar-dtvvar-methods`), 15

`summary.manmetavar.metavar`
 (`manmetavar-metavar-methods`),
 17

`summary.manmetavar.mplus`
 (`manmetavar-mplus-methods`), 19

`summary.manmetavar.naive`
 (`manmetavar-metavar-methods`),
 17

`vcov.manmetavar.dtvvar`
 (`manmetavar-dtvvar-methods`), 15

`vcov.manmetavar.metavar`
 (`manmetavar-metavar-methods`),
 17

`vcov.manmetavar.mplus`
 (`manmetavar-mplus-methods`), 19

`vcov.manmetavar.naive`
 (`manmetavar-metavar-methods`),
 17